Non-Ergodic Mess3 Mixtures: How Transformers Represent Hierarchical Belief Structure

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Abstract

We train a small transformer on a non-ergodic dataset of Mess3 processes with different parameters, where each sequence comes from one ergodic component. The model must solve a hierarchical inference problem: identifying the component (meta-synchronization) and tracking belief states within it. We derive geometric predictions connecting this to the Factored World Hypothesis, verify Sierpinski gasket recovery ($R^2 = 0.989$), and analyze synchronization dynamics across positions and layers.

1 Introduction

Language models face non-ergodic data where different documents constitute distinct statistical regimes. In-context learning requires identifying the active regime before predicting optimally. We study this in a controlled setting using Mess3 processes (Marzen and Crutchfield, 2017; Shai et al., 2024), 3-state nonunifilar HMMs whose belief geometry forms a Sierpinski gasket.

By constructing a non-ergodic training set where each sequence comes from one of K Mess3 parameterizations, we create a minimal in-context learning problem. This connects to the Factored World Hypothesis (Shai et al., 2026): component identity and within-component belief could be represented in orthogonal subspaces with dimensionality $(K-1)+2 = K+1$, rather than the joint $3K-1$. However, the joint transition operator is block-diagonal (not tensor-product), creating an intermediate case.

Contributions. (1) We formalize the non-ergodic mixture as hierarchical belief inference and derive geometric predictions (§4). (2) We verify single-process gasket recovery ($R^2 = 0.989$) and analyze the mixture against the Bayes-optimal predictor (§5). (3) We analyze synchronization dynamics: when and where the model identifies the active component (§6).

2 Background

Mess3 process. A 3-state nonunifilar HMM emitting $z \in \{0, 1, 2\}$, parameterized by emission clarity α and state persistence x, with $\beta = (1-\alpha)/2$, $y = 1 - 2x$. The labeled transition matrices $T^{(z)}_{ij} = P(\text{emit } z, \text{go to } j \mid \text{state } i)$ are given in Appendix A. Key properties: nonunifilar (optimal prediction requires belief tracking), belief states form a Sierpinski gasket in the 2-simplex, and the spectral gap $\zeta = 1-3x$ controls belief convergence rate (Riechers and Crutchfield, 2018).

Belief geometry in transformers. Shai et al. (Shai et al., 2024) showed transformers linearly represent belief states $\boldsymbol{\eta}^{(z_{1:t})} = \boldsymbol{\eta}^{(\varnothing)} T^{(z_1)} \cdots T^{(z_t)}/\text{norm}$ in their residual stream. Piotrowski et al. (Piotrowski et al., 2025) showed this is a constrained Bayesian approximation: $r_1(z_{1:d}) = \boldsymbol{\pi} + \sum_{s=1}^{d} (\boldsymbol{\pi} T^{z_s} T^{d-s} - \boldsymbol{\pi})$, where corrections decay as ζ^{d-s}.

Factored World Hypothesis. Transformers have an inductive bias toward factored representations in orthogonal subspaces (Shai et al., 2026). For N conditionally independent factors, the factored representation requires $\sum_n (d_n - 1)$ dimensions vs. $\prod_n d_n - 1$ (joint). This persists even when it sacrifices fidelity.

ICL as source discrimination. Non-ergodicity forces hierarchical inference: identify the source, then predict within it. This is structurally identical to in-context learning. ICL and activation steering both modify the model's belief state, operating additively in log-belief space.

3 Non-Ergodic Mess3 Mixture

3.1 Construction

We construct K Mess3 processes with distinct parameters $\{(\alpha_k, x_k)\}_{k=1}^{K}$. Each training sequence is

generated entirely by one component chosen uniformly; the model observes only tokens, not the component identity.

We choose components with well-separated entropy rates H_k and spectral gaps $\zeta_k = 1 - 3x_k$:

	α	x	H_k (nats)	ζ_k
C0	0.95	0.03	0.41	0.91
C1	0.85	0.05	0.69	0.85
C2	0.65	0.15	1.03	0.55

Table 1: Mess3 components. H_k: entropy rate (minimum CE). ζ_k: spectral gap (belief convergence rate). C0 is highly predictable with slow mixing; C2 is nearly uniform with fast mixing.

3.2 Relevance to Language Models

This structure mirrors natural language in three ways. **ICL as source discrimination**: the model must use early tokens to identify the generating process, analogous to topic/style adaptation. **Covariate shift**: different components have different token distributions, requiring in-context adaptation. **Hierarchical latent structure**: component identity is a slow variable (constant per sequence), hidden state is fast (evolves per token), mirroring the multi-scale structure of language.

3.3 Experimental Setup

Model. GPT-2 style decoder-only transformer via TransformerLens (Nanda and Bloom, 2022). Phase 1 (single Mess3): 3 layers, $d_{\text{model}} = 64$, 4 heads. Phase 2 (mixture): 4 layers, $d_{\text{model}} = 128$, 4 heads. Pre-norm, learned positional embeddings, vocabulary $\{$BOS, 0, 1, 2$\}$.

Training. Next-token CE loss, Adam (lr $= 5 \times 10^{-4}$, no weight decay). Fresh batches of 8192 sequences generated on-the-fly. Context: 16 tokens + BOS. Phase 1: 10K steps, Phase 2: 20K steps.

Baselines. Bayes-optimal CE at each position via Monte Carlo: both *oracle* (knows component) and *Bayesian* (infers component from observations).

Analysis. (1) CEV via PCA, (2) linear regression to belief states (R^2), (3) meta-belief regression, (4) per-position/layer probing for synchronization dynamics, (5) attention patterns.

4 Pre-Registered Predictions

4.1 Mathematical Derivation

The joint state space $(k, s) \in \{1, \ldots, K\} \times \{1, 2, 3\}$ has $3K$ states. Since component identity k is fixed within each sequence, the predictive vector factorizes:

$$P(k, s \mid z_{1:t}) = P(k \mid z_{1:t}) \cdot P(s \mid k, z_{1:t}) \quad (1)$$

The meta-belief $P(k \mid z_{1:t})$ is the posterior over components; $P(s \mid k, z_{1:t})$ is the within-component belief. The joint transition operator is block-diagonal $T^{(z)} = \bigoplus_k P(k) T_k^{(z)}$, which is *not* a tensor product, so the standard FWH does not directly apply. However, k is static, so after sufficient context $P(k \mid z_{1:t})$ concentrates on the true component, reducing to the single-component case.

The meta-belief evolves as $P(k \mid z_{1:t}) \propto P(k) \prod_{\tau=1}^{t} P(z_\tau \mid z_{1:\tau-1}, k)$, where concentration rate depends on the KL divergence between component distributions.

4.2 Geometric Predictions

Prediction 1: Dimensionality. Under factored representation: $(K-1)$ dims for the meta-belief simplex + 2 dims for within-component belief = $K+1$ total. For $K = 3$: factored $= 4$ dims, joint $= 8$ dims, separated gaskets ≈ 6 dims.

Prediction 2: Context-position dependence. At $t = 1$, meta-belief is near-uniform: geometry is a superposition of K gaskets. As t increases, meta-belief sharpens, geometry resolves into component-specific clusters. Per-position loss should decrease toward the Bayes-optimal rate, governed by inter-component KL divergence.

Prediction 3: Layer dependence. Embedding: 3 token vertices. Early layers: component discrimination emerges (meta-synchronization). Later layers: within-component gasket structure sharpens. This ordering reflects computational dependency: you must identify the process before applying the correct transition matrices.

4.3 Geometric Intuition

Three conceivable geometries: (1) **Separated gaskets**: K distinct gaskets with large offsets ($\sim 2K$ dims). (2) **Factored**: component identity in $(K-1)$D, shared 2D belief subspace with

superimposed gaskets ($K+1$ dims). (3) **Conditional**: component-specific 2D belief subspaces ($2K+K-1$ dims).

We predict geometry 1 or 2. Geometry 2 would extend the factoring inductive bias beyond conditionally-independent processes. We expect the model may initially learn geometry 1, transitioning toward 2 over training as in (Shai et al., 2026).

5 Results

5.1 Phase 1: Single Mess3 Verification

Before analyzing the non-ergodic mixture, we verify that our setup reproduces the known result: a transformer trained on a single Mess3 process ($\alpha = 0.85$, $x = 0.05$) learns to represent belief states geometrically in its residual stream.

Training. The model (3 layers, $d_{\text{model}} = 64$) was trained for 10K steps. The training loss converged to ~ 0.81 nats, compared to the entropy rate $H = 0.69$ nats and the uniform baseline $\ln 3 = 1.10$ nats. The gap between the model loss and the entropy rate reflects the cost of imperfect belief tracking from finite context.

Belief regression. Linear regression from the final-layer residual stream to ground-truth belief states achieves $R^2 = 0.989$ (RMSE $= 0.043$). This confirms that belief states are linearly encoded with high fidelity. The R^2 improves monotonically across layers: $L0 = 0.984$, $L1 = 0.987$, $L2 = 0.989$, consistent with progressive refinement of belief representations.

Geometry. Figure 1 shows the ground-truth Sierpinski gasket alongside the geometry recovered by linear regression from the residual stream. The fractal structure is clearly visible in both, confirming that the transformer has learned to represent the belief state geometry.

Dimensionality. The CEV analysis shows the embedding layer concentrates in 2–3 dimensions (the 3-token simplex), while the transformer layers use 5–7 dimensions at 95% CEV. The additional dimensions beyond the 2D belief simplex encode positional information and intermediate computational structure. Over training, the effective dimensionality decreases monotonically from 17 to 7 (Figure 2).

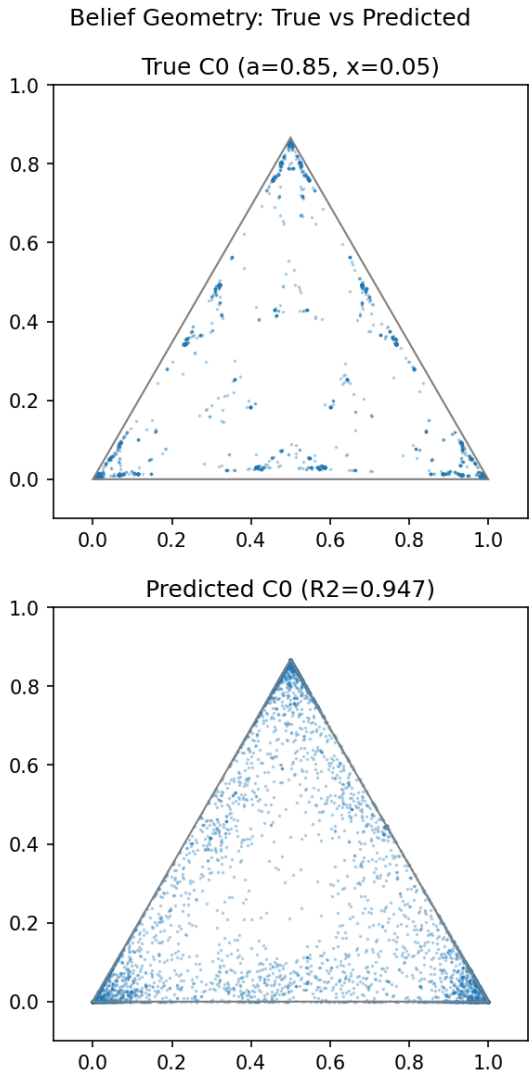

Figure 1: Single Mess3 ($\alpha = 0.85$, $x = 0.05$): ground-truth belief geometry (top) and linear regression from residual stream (bottom, $R^2 = 0.947$ at last position only). The Sierpinski gasket structure is clearly recovered.

5.2 Phase 2: Non-Ergodic Mixture

[Results to be filled after Phase 2 experiment completion. Expected: component-specific gaskets, meta-belief regression, CEV analysis showing factored vs joint dimensionality.]

5.2.1 Training Dynamics

5.2.2 Dimensionality Analysis

5.2.3 Belief Regression

5.2.4 Per-Position Loss vs Bayes-Optimal

6 Additional Analysis: Synchronization Dynamics

We analyze *synchronization dynamics*: how the model identifies the active component and when

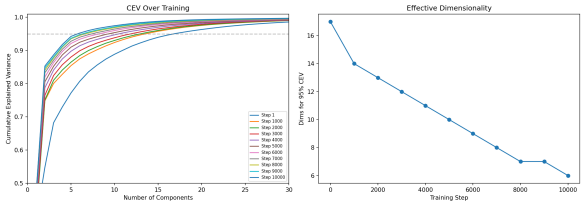

Figure 2: Left: CEV curves at different training steps, showing progressive concentration. Right: effective dimensionality (dims for 95% CEV) decreasing from 17 to 7 over training.

this precedes belief refinement.

Motivation. The model faces two tasks with different computational requirements: meta-synchronization (comparing accumulated statistics via likelihood ratios in log-probability space) and within-component belief tracking (constrained Bayesian updates (Piotrowski et al., 2025)). We probe each independently.

6.1 Position-wise probing

At each context position t, we train linear probes for (a) meta-belief $P(k \mid z_{1:t})$ and (b) within-component belief $P(s \mid k, z_{1:t})$ from the final-layer residual stream.

[Results to be filled: R^2 curves by position.]

6.2 Layer-wise probing

At each layer, we independently probe for component identity and belief state, revealing the order of representation construction.

[Results to be filled: per-layer R^2.]

6.3 Attention patterns

We examine whether specific heads specialize for meta-synchronization (broad attention for evidence gathering) vs. belief updating (local attention).

[Results to be filled.]

6.4 Connection to ICL

This provides a concrete mechanistic account: the model first identifies the source, then conditions predictions on it. The per-position loss curve against the Bayes-optimal baseline quantifies the "synchronization gap" from imperfect source identification.

7 Discussion

Non-ergodicity forces hierarchical inference. The model cannot achieve low loss without first identifying the source. This mirrors how LLMs must identify the regime (topic, style, task) before predicting well.

Block-diagonal vs tensor-product. The joint operator is block-diagonal, not tensor-product, so standard FWH does not directly apply. Yet we observe partial factoring, suggesting the factoring inductive bias extends beyond conditionally-independent processes and acts as a "representational attractor."

Residual stream as belief space. The residual stream has no privileged basis; geometric arrangement encodes prediction similarity. Factored structure means the model discovered that source identity and within-source state are independent dimensions of prediction, allocating orthogonal subspaces.

Limitations. Small model (3–4 layers); static components (no drift/switching); same token alphabet across components. Extensions: slowly drifting parameters, shared/component-specific tokens, connections to induction heads (Olsson et al., 2022).

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A Mess3 Transition Matrices

With $\beta = (1-\alpha)/2$ and $y = 1 - 2x$:

$$T^{(0)} = \begin{pmatrix} \alpha y & \beta x & \beta x \\ \alpha x & \beta y & \beta x \\ \alpha x & \beta x & \beta y \end{pmatrix} \tag{2}$$

$$T^{(1)} = \begin{pmatrix} \beta y & \alpha x & \beta x \\ \beta x & \alpha y & \beta x \\ \beta x & \alpha x & \beta y \end{pmatrix} \tag{3}$$

$$T^{(2)} = \begin{pmatrix} \beta y & \beta x & \alpha x \\ \beta x & \beta y & \alpha x \\ \beta x & \beta x & \alpha y \end{pmatrix} \tag{4}$$

The net transition matrix is:

$$T = T^{(0)} + T^{(1)} + T^{(2)} = \begin{pmatrix} 1-2x & x & x \\ x & 1-2x & x \\ x & x & 1-2x \end{pmatrix} \tag{5}$$

with uniform stationary distribution $\boldsymbol{\pi} = [1/3, 1/3, 1/3]$ and non-trivial eigenvalue $\zeta = 1 - 3x$.

The emission probability $P(z \mid s) = \sum_j T^{(z)}_{sj}$ gives the entropy rate:

$$H = -\sum_{s=1}^{3} \frac{1}{3} \sum_{z=0}^{2} P(z|s) \log P(z|s) \tag{6}$$

B Bayes-Optimal Loss Derivation

The Bayes-optimal loss at position t is the cross-entropy achieved by a predictor maintaining exact belief states:

$$\mathrm{CE}(t) = \mathbb{E}_{z_{1:t}} \left[-\log P(z_t \mid z_{1:t-1}) \right] \tag{7}$$

where $P(z_t \mid z_{1:t-1}) = \boldsymbol{\eta}^{(z_{1:t-1})} T^{(z_t)} \mathbf{1}$.

For the non-ergodic mixture, we compute two baselines:

- **Oracle**: knows the component, uses component-specific transition matrices.
- **Bayesian**: must infer the component, uses the mixture prediction $P(z_t \mid z_{1:t-1}) = \sum_k P(k \mid z_{1:t-1}) \cdot P(z_t \mid z_{1:t-1}, k)$.

The gap between oracle and Bayesian baselines quantifies the cost of not knowing the component identity.

C Code

All code is available at: `[GitHub repo URL]`. The implementation uses PyTorch with TransformerLens for activation extraction. Key files:

- `src/mess3.py`: Mess3 process generation and belief computation
- `src/bayes_optimal.py`: Bayes-optimal loss computation
- `src/sync_analysis.py`: Synchronization dynamics analysis
- `run.py`: Main experiment script